

Kaleidoscopic Display: easy-to-make light-field display

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CCS CONCEPTS

• Hardware → Displays and imagers.

KEYWORDS

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1 INTRODUCTION

Light-field display is one type of 3-dimensional display that requires no special glasses. A typical method of this display is integral photography[Lippmann 1908]. A lens array is mounted on an ordinary 2-dimensional display in this method to provide the parallax. Although this structure is simple and suitable for mass production, the initial cost of lens array tends to be more expensive for prototyping or small lot production. On the other hand, a projector-array-based method is often used in large-scale light-field display[Kawakita et al. 2018]. Although the size of the projection type display is easy to be scaled up, the multi-projection system in the display tends to be larger and more expensive because many projectors are required to achieve smooth parallax.

In this article, we propose a multi-projection-type light-field display by using only a single projector. This system which is small and simple, employs kaleidoscope-like optics to generate a virtual projector array. Many virtual projectors can be generated with multi-reflections on a small number of mirrors. Since this method consists of a small number of readily available components, this method is suitable for experimental purposes or a small lot production. The proposed method is highly scalable as it can be adapted into a small display with high ray density or a large display for multiple observers.

The methods using mirrors to create virtual projectors have been explored as a part of the capturing system with illumination modulation. These methods are classified into the single-reflection type or the multi-reflection type. In the single-reflection type system, each mirror corresponds to each virtual projector[Mukaigawa et al. 2011]. Besides, by applying the concept of mirror-array, this method is also used for light-field display[Chen et al. 2019]. On the other

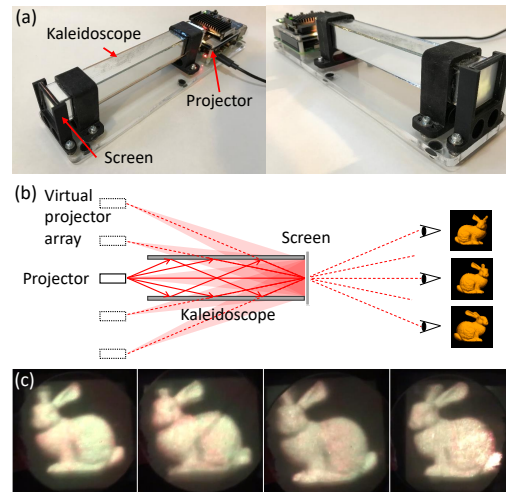


Figure 1: (a) A prototype of the kaleidoscopic display. (b) Structure of the proposal. (c) A 3-dimensional image displayed by the prototype using condenser lens on a screen.

hand, the multi-reflection type system employs kaleidoscope-like optics, and it was used for bidirectional texture function acquisition[Han and Perlin 2003]. However, the multi-reflection type method has not been applied for a light-field display to our best knowledge. Our study is novel research on the light-field display that employs kaleidoscope-like optics. The proposal is low-cost and easy to build because it requires a much smaller number of mirrors than the single-reflection type.

2 PROPOSAL

The proposal consisted of a kaleidoscope, a projector, and a rear-projection type screen(Fig.1(b)). The rays generated by the projector are reflected in the kaleidoscope and projected onto the screen from various angles; in other words, the mirrors create a virtual projector array. The observers can perceive different images according to the viewpoint by using a weak diffusive translucent screen.

The display property can be adjusted with the kaleidoscope design. For example,

- A longer kaleidoscope can increase the number of reflections. It generates more virtual projectors, and it results in smoother parallax. (Fig.2(a))
- A tapered kaleidoscope can change the viewing angle of the display by changing the projector alignment. (Fig.2(b))

On the other hand, since the projection image of the real projector is divided into small images of virtual projectors, the overall number of rays is the same as the number of pixels in the real projector. Thus,

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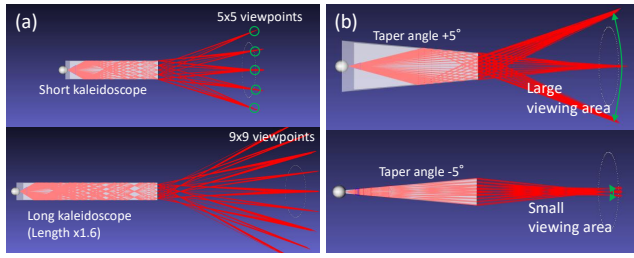


Figure 2: Light-ray preview of (a) different length and (b) different taper angle of kaleidoscope.

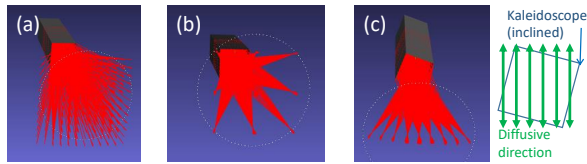


Figure 3: The light-ray preview of (a) weak diffusive screen, (b) condenserlens (3×3 viewpoints) and (c) condenserlens + anisotropic diffuser (9×1 viewpoints).

there is a tradeoff between apparent resolution and the number of viewpoints similar to the integral-photography-based method.

There are several options on the rear-projection screen. The weak diffusive screen (Fig.3(a)) is the cheapest option. By adding a condenser lens on the screen (Fig.3(b)), the lens concentrates rays into the viewing zone, and the pixels are used effectively. In these two options, the generated projector array is two-dimensional, and that achieves both vertical and horizontal parallax. The anisotropic diffuser generates a one-dimensional array of the projector (Fig.3(c)). By using an anisotropic diffuser in the inclined direction from the kaleidoscope, a high-density projector array can be generated, and then smooth horizontal only parallax is achieved.

3 IMPLEMENTATION

We prototyped the display system to confirm the validity and feasibility of the proposal (Fig.1(a)). The prototype consisted of a MEMS laser scanning projector with a resolution of 1280×720 pixels, four surface reflection mirrors, and the 3D printed screen holder. The kaleidoscope is non-tapered, and the projection distance is 200 mm approximately. The display area is 20 × 20 mm, and the number of virtual projectors is 7×3.

To calibrate geometrical error, we projected calibration patterns and captured the images on the screen in advance. Then we estimate optical parameters from the captures to create the model of the projection system. Finally, the projection image (Fig.4(a)) is created by ray tracing on the estimated system model.

Fig.1(c) shows the result using a stacked screen that consists of a polypropylene sheet as a diffuser, an aluminized film as an attenuator, and a condenser lens of 100 mm focal length. Virtual viewpoints are generated in the interval of 20 mm on a plane of 200 mm away from the screen. We also experimented using a screen without a condenser lens and a screen of an anisotropic diffuser (Fig.4(b)).

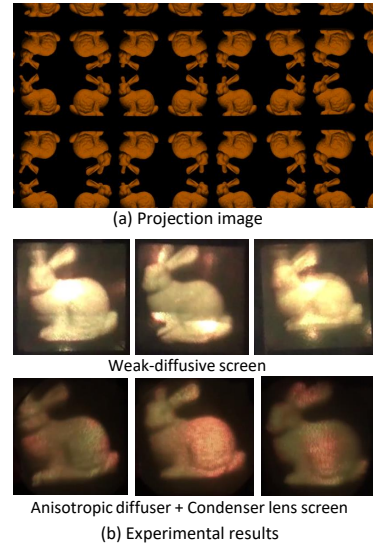


Figure 4: (a) A projection image (b) Experimental results in different screens

Without a condenser lens, the displayed image is brightened partially because of the reflected images of a projector aperture. The anisotropic diffusive screen could generate smoother parallax. As a result, we confirmed that observers could perceive 3-dimensional images with a correct parallax from our prototype in each screen setup.

4 LIMITATIONS AND FUTURE WORKS

A projector with a small depth-of-field can cause problems, since both screen and kaleidoscope should be in focus. The influences of depth-of-field should be analyzed toward using DLP/LCoS based projectors.

In this article, the virtual projector array is generated from a single projector. To increase the spatial resolution, multiple projector or time-multiplexing can be employed with a kaleidoscope.

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